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(54) **HYDROGENATION OF ACETYLENES IN A HYDROCARBON STREAM**

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CPC C10G 70/041; C10G 51/06; C10G 2300/103;
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See application file for complete search history.

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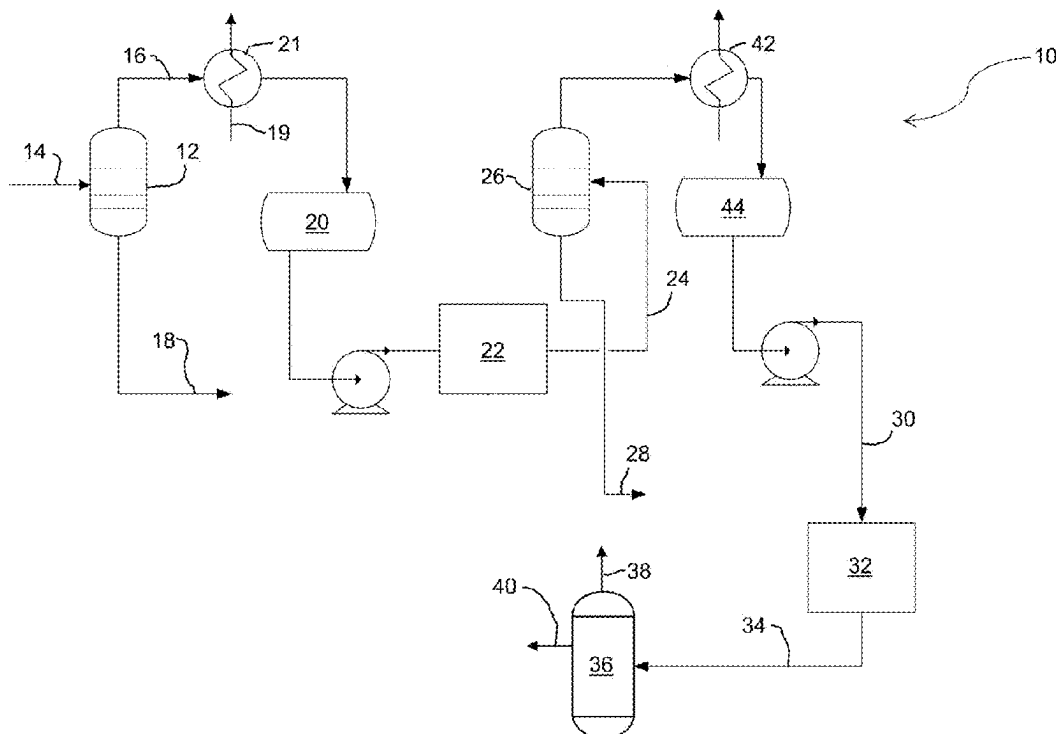
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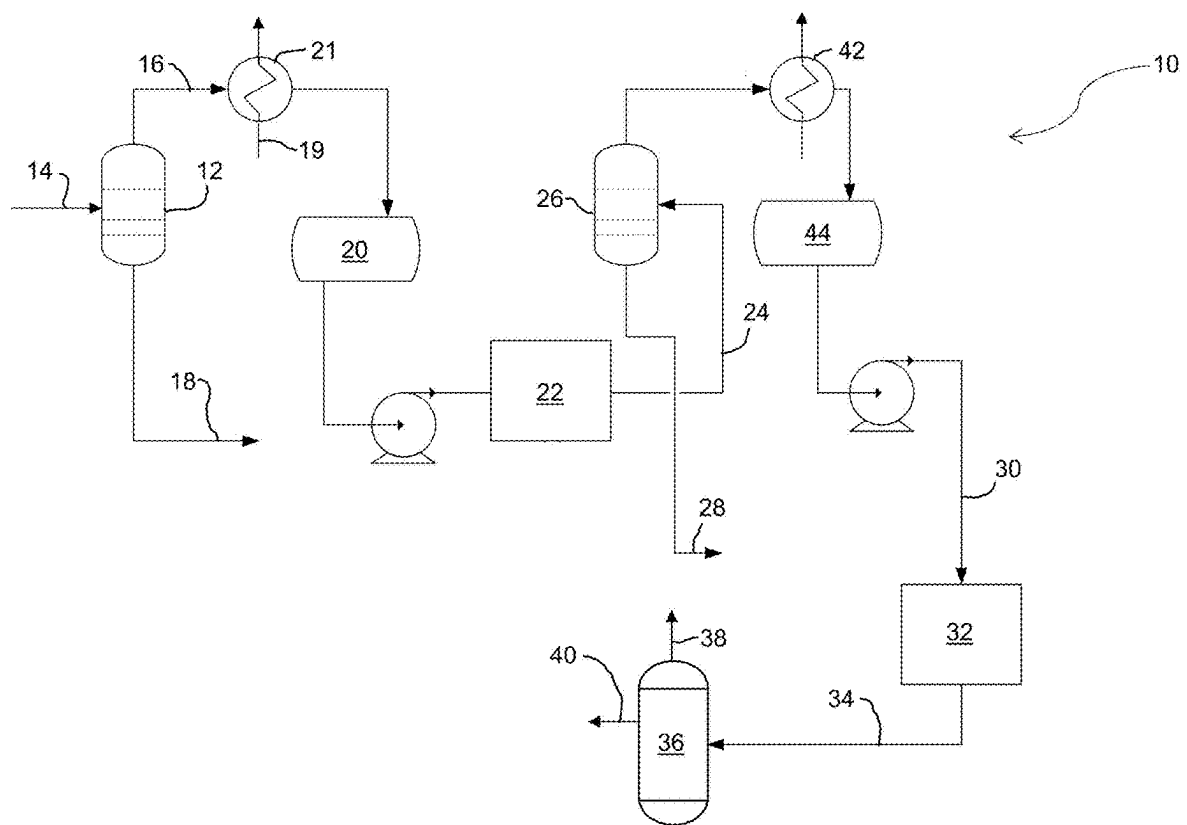
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(57) **ABSTRACT**

A system for hydrogenation C₃ and C₄ acetylenes contained
within a hydrocarbon stream generated in a stream cracker
unit where a debutanizer is placed upstream of a depro-
panizer for more economical processing of the hydrocarbon
stream to produce lighter hydrocarbons, where the system
requires only one stripper tower downstream of hydrogena-
tion to remove residual hydrogen.

4 Claims, 1 Drawing Sheet





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HYDROGENATION OF ACETYLENES IN A HYDROCARBON STREAM

CROSS REFERENCE TO RELATED APPLICATION

This application is a divisional of U.S. application Ser. No. 17/729,975, filed Apr. 26, 2022, which claims priority to U.S. Provisional Patent Application having Ser. No. 63/180,296 filed on Apr. 27, 2021, the contents of which are hereby incorporated by reference in its entirety.

TECHNICAL FIELD

The present invention relates to systems for hydrogenation of C_4 acetylenes, such as vinyl and ethyl acetylenes, and C_3 acetylenes, such as methyl acetylene and propadiene, contained within a hydrocarbon stream being processed in a steam cracking process, and more particularly relates to such steam cracking processes required to meet product specifications on respective acetylenes.

BACKGROUND

In a conventional steam cracking process for the production of lighter hydrocarbons, a depropanizer, which separates a C_3 acetylene-rich hydrocarbon stream from C_4 acetylene-rich hydrocarbon stream, is placed upstream from a debutanizer. The C_4 acetylene-rich hydrocarbon stream leaving the depropanizer is sent to a debutanizer in which the C_4 acetylene-rich hydrocarbon is separated from the C_5 - C_{10} containing pyrolysis gas stream. The C_4 acetylene-rich hydrocarbon overhead of the debutanizer is then sent to a hydrogenation reactor in which it is hydrogenated over a selective catalyst. The C_3 acetylene-rich hydrocarbon overhead of the depropanizer is independently sent to a separate hydrogenation reactor to selectively hydrogenate the C_3 acetylene-rich hydrocarbon stream.

In this typical design, two separate stabilizing/stripper columns are required—one to vent off residual unreacted H_2 (hydrogen) from the C_3 hydrogenation reactor product stream and one to vent off residual unreacted H_2 (hydrogen) from the C_4 hydrogenation reactor product stream.

One prior process is seen in U.S. Pat. No. 5,090,977 to Exxon Chemical Patents Inc. that discloses a process sequence for treating cracked gases of heavy feedstocks which preferentially produces propylene to the exclusion of propane, butanes and butenes. The process eliminates the need for a depropanizer with the attendant savings in capital and operating costs. In lieu of a conventional C3 splitter, the process features a depropylenizer, i.e. a distillation tower designed to separate propylene from propane, butanes and butenes. A hydrogenation unit to eliminate contaminants can be placed upstream of the depropylenizer or the depropylenizer can be split into two sections with the hydrogenation unit located between the two sections.

Integration of gas oil and light olefin catalytic cracking zones with a pyrolytic cracking zone to maximize efficient production of petrochemicals is disclosed in U.S. Pat. No. 7,128,827 to Kellogg, Brown & Root LLC. Integration of the units in parallel allows production of an overall product stream with maximum ethylene and/or propylene by routing various feedstreams and recycle streams to the appropriate cracking zone(s), e.g. ethane/propane to the steam pyrolysis zone and C_4 C_6 olefins to the light olefin cracking zone. This integration enhances the value of the material balances produced by the integrated units.

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Additionally, U.S. Pat. No. 7,294,749 to Kellogg, Brown & Root LLC describes a low-pressure olefins recovery process and plant. The feed gas 300 is compressed 302, 304 and distilled 310 at a primary distillation pressure. The overhead stream 312 is chilled 318 at a pressure less than 30 kg/cm² (430 psia) to partially condense the overheads. The primary distillation tower 310 is refluxed with at least a portion of the condensate 320. The overhead vapor is further chilled 318 and partially condensed and the condensate 322 is fed to a demethanizer 324. The remaining vapor 326 is cooled in a cold section 328 and the resultant liquid is phase-separated 330 and expanded 331, 334 to provide refrigeration for the cold section. The expanded vapor 332 from the cold section is recycled to the process gas compressor. The bottoms streams 338, 342 from the primary distillation zone and the demethanizer are fractionated into respective streams consisting essentially of ethylene 356, ethane 358, propylene 364, propane 366, C_4 's 346, and C_{5+} 348.

However, there are significant capital and maintenance costs with building a steam cracking unit in which the system for hydrogenating acetylenes has this two stabilizing/stripper column design, which can create an economic barrier to building systems for oil and gas refining.

Thus, there is a need to lower capital costs associated with the production of lighter hydrocarbons and the hydrogenation of acetylenes in hydrocarbon streams, in particular.

SUMMARY

There is provided, in one non-limiting embodiment, a system for hydrogenating acetylenes in a hydrocarbon stream, the system including a depropanizer, a debutanizer, a C_4 acetylene hydrogenation reactor, a C_3 acetylene hydrogenation reactor, and a stripper column, where the depropanizer is located downstream of the debutanizer. In a different non-restrictive aspect, there is no stripping column additional to the stripper column for venting off residual unreacted H_2 from the crude butadiene stream.

In another non-limiting embodiment, there is provided a process for selective hydrogenation of acetylenes including directing a C_3 - C_{10} hydrocarbon stream to a debutanizer, selectively hydrogenating C_4 acetylenes in the condensed C_3 - C_4 overhead stream in a C_4 acetylene hydrogenation reactor to give a reactor outlet stream, directing the reactor outlet stream to a depropanizer downstream from the debutanizer, drawing a top stream from the depropanizer containing a C_3 mixture comprising methyl acetylene and propadiene, drawing a bottom stream from the depropanizer containing a crude C_4 mixture, directing the C_3 mixture to a methyl acetylene and propadiene (MAPD) reactor and selectively hydrogenating the methyl acetylene and propadiene to give a crude propylene stream, and directing the crude propylene stream to a stripper column. In a different non-restrictive aspect, the stripper column is the only stripper column and there is an absence of a stripper column downstream of the C_4 acetylene hydrogenation reactor.

In a different non-limiting embodiment there is provided an ethylene plant including a system for hydrogenating acetylenes in a hydrocarbon stream, the system including a depropanizer, a debutanizer, a C_4 acetylene hydrogenation reactor, a C_3 acetylene hydrogenation reactor, and a stripper column, where the depropanizer is located downstream of the debutanizer. In a different non-restrictive aspect of the ethylene plant, there is no stripping column additional to the stripper column for venting off residual unreacted H_2 from the crude butadiene stream.

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There is additionally provided an upgraded hydroprocessed product produced by the selective hydrogenation process herein that comprises from about 0.5 independently to about 2 wt % hydrogenated C₄ acetylenes; alternatively from about 0.9 independently to about 1.6 wt % hydrogenated C₄ acetylenes. Additionally, the upgraded hydroprocessed product has from about 20 wt % independently to about 60 wt % hydrogenated 1,3-butadiene; alternatively from about 40 independently to about 50 wt % hydrogenated 1,3-butadiene. The upgraded hydroprocessed product may also have from about 20 wt % independently to about 60 wt % of hydrogenated C₃ hydrocarbons; alternatively from about 40 wt % independently to about 50 wt % of hydrogenated C₃ hydrocarbons.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates an example diagram of the system for hydrogenating acetylenes in a hydrocarbon stream as described herein.

DETAILED DESCRIPTION

It has been discovered that having a system for hydrogenating acetylenes in a stream cracker in which the debutanizer is located upstream from the depropanizer can eliminate the need for two separate stabilizing/stripping columns to vent off residual unreacted H₂ (hydrogen) present in the crude butadiene and crude propylene product streams. In other words, the present system and process switches the sequence of depropanizer and debutanizer where in debutanizer comes first and depropanizer is placed downstream of debutanizer. This is done because both C₄ acetylenes (vinyl acetylene, ethyl acetylene) and C₃ acetylenes (methyl acetylene, propadiene) need to be hydrogenated to meet crude butadiene and crude propylene product specifications on respective acetylenes. As noted, in the conventional sequence, this requires two separate stabilizing columns—one for crude butadiene and one for crude propylene to vent off the residual hydrogen unreacted in the hydrogenation reactor. These two separate columns add to the capital cost of the project. Thus, this system and process saves one tower which is significant cost saving.

A gas cracker typically includes an ethane feed (or a mix of ethane and propane) to a Pyrolysis System. The cracked gas passes to a Quench and Condensate Recovery System where it is cooled. The cooled crack gas feeds the Compression, Acid Gas Removal and Drying Systems. The compressed cracked gas feeds a Deethanizer and Acetylene Reactor Systems where the C₂ and lighter components are separated from the C₃ and heavier components. The C₃ and heavier components make up three products that have value: a propylene-rich mixed C₃s product, a crude butadiene product, and a pyrolysis gasoline product.

Producing the crude butadiene product normally would have the C₃ and heavier components feed a Depropanizer, with the bottoms then feeding a Debutanizer to produce a mixed C₄s product on the Debutanizer overhead. The Debutanizer overhead can then be treated in a C₄ Acetylene Reactor System provided to upgrade the crude butadiene product as the reduction or elimination of acetylenes can yield higher market value for the product stream or it could result in savings in a butadiene unit that exceed the cost of the C₄ Acetylene Reactor System. However the C₄ Acetylene Reactor System adds light end contaminants that would need to be removed by an additional Stripper System.

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The hydrogenation of C₃ and C₄ acetylenes contained within a hydrocarbon stream being processed in a steam cracker unit stream may be accomplished, in one non-limiting embodiment, by the system shown in FIG. 1 and by the process described herein.

Referring to FIG. 1 and the overall system for hydrogenating acetylenes in a hydrocarbon stream 10, a debutanizer 12 receives a feed containing a mixture of C₃-C₁₀ range hydrocarbons 14. In one embodiment, the C₄ and C₃ hydrocarbons in the stream may be a mixture of saturates, olefins, diolefins, and acetylenes. Example feed conditions of the C₄ acetylenes and hydrogen (H₂) feed are provide in Table Y below along with operating conditions.

TABLE Y

Feed conditions		
Operating conditions		
Pressure, kg/cm ² (psia)	24 (340)	
Temperature (inlet), F. (C.)	93 (34)	
Feed, lb/hr (kg/hr)	1 (0.45)	
H ₂ , lb/hr (g/hr)	0.0007 (0.318)	
Components	C ₄ acetylenes reactor feed wt %	H ₂ wt %
Hydrogen	0.01	100
Ethane	0.04	
Methyl Acetylene	0.6	
Propadiene	0.4	
Propylene	41	
Propane	7	
Vinyl Acetylene	0.7	
1-Butyne	0.1	
1,3-Butadiene	38	
1-Butene	4	
Cis-2-Butene	1	
Trans-2-Butene	1	
Isobutylene	0.2	
n-Butane	6	
3-Methyl-1-Butene	0.1	
Total	100	100

The debutanizer overhead stream 16 is a mixture of C₃ and C₄ hydrocarbons, while the debutanizer bottoms stream 18 is a pyrolysis gasoline product stream containing C₅-C₁₀ hydrocarbons. It will be appreciated that the pressure and temperature of debutanizer 12 should be set in a manner in which the hydrocarbons in the debutanizer overhead stream 16 can be condensed using available economical refrigeration mediums. In one non-restrictive embodiment, the hydrocarbons are condensed at a pressure of about 50 psig (0.3 MPa). If condensed at this pressure, it will be appreciated that, since the debutanizer overhead stream 16 comprises a mixture of C₃ and C₄ hydrocarbons, a refrigeration medium 19 may be used in debutanizer condenser 21, and that the temperature of the debutanizer overhead stream 16 may be in the range of about 40-50° F. (about 4-10° C.) vs. conventional temperature of 100° F. (about 38° C.). The refrigeration medium 19 may be a propylene stream from a steam cracker (not shown).

Continuing reference to FIG. 1, the debutanizer overhead stream 16, which is a mixture of C₃ and C₄ hydrocarbons in the form of saturates, olefins, and/or diolefins, after being directed through debutanizer reflux drum 20, is pumped to C₄ acetylene selective hydrogenation reactor 22 where C₄ acetylenes are selectively hydrogenated. In one non-limiting embodiment, the proportion of C₃ hydrocarbons in debutanizer overhead stream 16 is at least 20 wt % C₃ hydrocar-

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bons and C₄ hydrocarbons including C₄ acetylenes and 1,3-butadiene. A molecular hydrogen-containing stream (not shown) introduced prior to the C₄ acetylene selective hydrogenation reactor 22 may have at least 35 wt % molecular hydrogen, alternatively at least 99 wt % molecular hydrogen, and in another non-limiting version may be substantially pure molecular hydrogen.

The temperature at the C₄ acetylene selective hydrogenation reactor 22 inlet is in a range of from about 50° F. (10° C.) independently to about 140° F. (about 60° C.) from start of run to end of run, alternatively from about 78° F. (25° C.) independently to about 95° F. (35° C.). As used herein with respect to a parameter range, the word “independently” means that any range endpoint may be used together with any other range endpoint to give an acceptable alternative range.

The C₃ and C₄ hydrocarbons should be around 99 percent in liquid phase. The C₃ and C₄ vaporization is not used to control the reactor temperature. The reactor 22's total pressure is at least that needed to maintain the C₃ and C₄ hydrocarbons in the liquid phase during the hydrogenation. In one non-limiting embodiment the pressure ranges from about 300 psia (about 2.1 MPa) independently to about 600 psia (about 4.9 MPa); alternatively from about 350 psia (about 2.5 MPa) independently to about 400 psia (2.8 MPa). Higher pressure is better for hydrogen solubility in feed and should be high enough to keep hydrocarbons in liquid phase through the C₄ acetylene selective hydrogenation reactor 22.

In one embodiment, the C₄ acetylene converter outlet components are:

Components	C4 acetylene converter outlet wt %
Hydrogen	0.01
Methane	0.00
Ethane	0.04
Methyl Acetylene	0
Propadiene	0
Propylene	41
Propane	7
Vinyl Acetylene	0
1-Butyne	0
1,3-Butadiene	37
1-Butene	5
Cis-2-Butene	1
Trans-2-Butene	1
Isobutylene	0
n-Butane	6
Cyclopentadiene	0
2-Methyl-1,3-Butadiene	0
Total	100

Selective hydrogenation is carried out by controlled injection of H₂ to hydrogenate C₄ acetylenes selectively on a catalyst fixed bed. Unreacted H₂ from effluent of reactor is sent to cracked gas compressor suction after cooling and separating in a separator drum

The upgraded C₃ and C₄ outlet stream 24 of the C₄ acetylene selective hydrogenation reactor 22 is sent to a depropanizer 26, which separates a bottoms crude butadiene product stream 28 from net depropanizer overhead stream 30. The depropanizer bottoms stream is the bottoms crude butadiene product stream 28, while the net depropanizer overhead stream 30 is a mixture of C₃ hydrocarbons, the latter which is passed through depropanizer condenser 42. The net depropanizer overhead stream 30 is then pumped from depropanizer reflux drum 44 to MAPD (methyl acetylene and propadiene) reactor 32 to selectively hydrogenate

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C₃ acetylenes, such as, without limitation, methyl acetylene and propadiene. After selective hydrogenation of C₃ acetylenes in the MAPD reactor 32, the upgraded mixed C₃ hydrocarbon stream 34 leaving the MAPD reactor 32 goes to a C₃ stripper column 36 where unreacted H₂ and lights 38 (typically comprising methane and trace C₂s) are vented out and propylene product 40 is withdrawn, in one form, as a side draw from the C₃ stripper column 36. In this non-limiting embodiment, it is appreciated that because the depropanizer 26 functions to separate lighter hydrocarbons from C₄ hydrocarbons in the stream, it is not necessary for the crude butadiene product stream 28 leaving the bottom of the depropanizer 26 to be sent to a separate stripping column to vent off residual unreacted H₂.

In one non-limiting embodiment, the hydrocarbons in the stream, feed containing a mixture of C₃-C₁₀ range hydrocarbons 14, may be a mixture of saturates, olefins, and acetylenes.

The hydrogenation catalyst(s) useful for selectively hydrogenating the C₄ acetylenes in the stream may include, but are not necessarily limited to, palladium-based catalysts, such as; palladium-on-alumina, copper-based catalysts, rhodium-based catalysts, and other such metal based catalysts. The hydrogenation catalyst(s) useful for selectively hydrogenating the C₃ acetylenes in the stream may include, but are not necessarily limited to, palladium-based catalysts, such as; palladium-on-alumina, copper-based catalysts, rhodium-based catalysts, and other such metal based catalysts.

It is appreciated that conversion of C₄ acetylenes can be targeted to meet the specifications for C₄ acetylenes in the product crude butadiene stream. This is accomplished by controlled injection of H₂ to hydrogenate C₄ acetylenes selectively. The reactor size, recycle rate, pressure, and temperature of reactor inlet may be designed to achieve or exceed the desired conversion of C₄ acetylenes and to decrease or even minimize conversion of the 1,3-butadiene and propylene present in the crude butadiene stream. In one exemplary embodiment, the desired run length may be in the range of about 2 months independently to about 12 months; alternatively from about 6 independently to about 9 months; and the range of space velocity (LHSV) may be in the range of 4 independently to about 20; alternatively from about 8 independently to about 16.

It will also be appreciated that any hydrogenation of C₃ acetylenes, such as methyl acetylene and propadiene, in this process, though not intended, is beneficial.

The processes and systems described herein may accomplish a variety of goals including, but not necessarily limited to:

hydrogenating C₄ acetylenes in mixed hydrocarbon streams in a manner that substantially maintains selectivity and conversion for hydrogenating the C₄ acetylenes;

hydrogenating C₄ acetylenes in mixed hydrocarbon streams in a manner that gives increased selectivity and/or conversion; and

hydrogenating C₄ acetylenes in mixed hydrocarbon streams with decreased hydrogenation of valuable hydrocarbons such as 1,3-butadiene and propylene.

The processes and systems described herein are considered effective and successful even if only one of these goals is accomplished, such as achieving substantially the selectivity and conversion of C₄ acetylenes in a mixed C₃/C₄ stream with a C₄ stream. The processes and systems may be considered even more effective if one or more of the other goals is/are also achieved.

In the foregoing specification, the invention has been described with reference to specific embodiments thereof. However, the specification is to be regarded in an illustrative rather than a restrictive sense. For example, hydrogenation reaction conditions and equipment, debutanizer and depropanizer conditions, catalysts, and composition and conditions of hydrocarbon and acetylene streams, hydrogen streams, falling within the claimed or disclosed parameters, but not specifically identified or tried in a particular example, are expected to be within the scope of this invention.

The present invention may be practiced in the absence of an element not disclosed. In addition, the present invention may suitably comprise, consist or consist essentially of the elements disclosed. For instance, the system may comprise, consist of, or consist essentially of: a depropanizer, a debutanizer, a C₄ acetylene-rich hydrogenation reactor, a methyl acetylene and propadiene (MAPD) reactor, and a stripper column, where the depropanizer is located downstream of the debutanizer.

In another non-limiting embodiment there may be provided a process for selective hydrogenation of acetylenes comprising, consisting essentially of, or consisting of directing a C₃-C₁₀ hydrocarbon stream to a debutanizer, condensing a debutanizer C₃-C₄ overhead stream, selectively hydrogenating C₄ acetylenes in the condensed C₃-C₄ overhead stream in a C₄ acetylene hydrogenation reactor to give a reactor outlet stream, directing the reactor outlet stream to a depropanizer downstream from the debutanizer, drawing a top stream from the depropanizer containing a C₃ mixture comprising methyl acetylene and propadiene, drawing a bottom stream from the depropanizer containing a crude C₄ mixture, directing the C₃ mixture to a methyl acetylene and propadiene (MAPD) reactor and selectively hydrogenating the methyl acetylene and propadiene to give a crude propylene stream, and directing the crude propylene stream to a stripper column.

Additionally, there may be provided an upgraded hydroprocessed product produced by the selective hydrogenation process herein that comprises from about 0.5 independently to about 2 wt % hydrogenated C₄ acetylenes; alternatively from about 0.9 independently to about 1.6 wt % hydrogenated C₄ acetylenes. Additionally, the upgraded hydroprocessed product has from about 20 wt % independently to about 60 wt % hydrogenated 1,3-butadiene; alternatively from about 40 independently to about 50 wt % hydrogenated 1,3-butadiene.

Furthermore there may be provided in another non-limiting embodiment an ethylene plant comprising, consisting essentially of, or consisting of a system for hydrogenating acetylenes in a hydrocarbon stream, the system comprising, consisting essentially of, or consisting of, a depropanizer, a debutanizer, a C₄ acetylene hydrogenation

reactor, a methyl acetylene and propadiene (MAPD) reactor, and a stripper column, where the depropanizer is located downstream of the debutanizer.

The words “comprising” and “comprises” as used throughout the claims, are to be interpreted to mean “including but not limited to” and “includes but not limited to”, respectively.

To the extent used herein, the word “substantially” shall mean “being largely but not wholly that which is specified.”

As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.

To the extent used herein, the term “about” in reference to a given parameter is inclusive of the stated value and has the meaning dictated by the context (e.g., it includes the degree of error associated with measurement of the given parameter).

To the extent used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

What is claimed is:

1. A process for selective hydrogenation of acetylenes comprising:

directing a C₃-C₁₀ hydrocarbon stream to a debutanizer; condensing a debutanizer C₃-C₄ overhead stream; selectively hydrogenating C₄ acetylenes in the condensed C₃-C₄ overhead stream in a C₄ acetylene hydrogenation reactor to give a reactor outlet stream; directing the reactor outlet stream to a depropanizer downstream from the debutanizer; drawing a top stream from the depropanizer containing a C₃ mixture comprising methyl acetylene and propadiene; drawing a bottom stream from the depropanizer containing a crude C₄ mixture; directing the C₃ mixture to a methyl acetylene and propadiene (MAPD) reactor and selectively hydrogenating the methyl acetylene and propadiene to give a crude propylene stream; and directing the crude propylene stream to a stripper column to vent off residual unreacted hydrogen.

2. The process of claim 1 where the stripper column is the only stripper column and in the absence of a stripper column downstream of the crude C₄ mixture.

3. The process of claim 1 where condensing a debutanizer C₃-C₄ overhead stream is conducted against a refrigeration medium.

4. The process of claim 1 where the C₄ acetylene hydrogenation reactor is operated at a temperature of from about 50° F. (10° C.) to about 140° F. (about 60° C.), and a pressure of from about 300 psia (about 2.1 MPa) to about 600 psia (about 4.9 MPa).

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